Chapter 2 - Modules

Flandre

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Exercise 1

Since $\gcd(m, n) = 1, \exists u, v \in \mathbb{Z}, \text{s.t.}$

$$mu + nv = 1$$

Now for $\forall x \otimes y \in (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z})$,

$$\begin{aligned} x \otimes y &= (mu + nv)(x \otimes y) \\ &= mu(x \otimes y) + nv(x \otimes y) \\ &= 0 \end{aligned}$$

$\square$

Exercise 2

Consider the exact sequence

$$\mathfrak{a} \xrightarrow{f} A \xrightarrow{g} A/\mathfrak{a} \to 0$$

By *Proposition 2.18*, we have the following exact sequence

$$\mathfrak{a} \otimes M \xrightarrow{f \otimes 1} A \otimes M \xrightarrow{g \otimes 1} (A/\mathfrak{a}) \otimes M \longrightarrow 0$$

Notice there exist trivial isomorphism $\mathfrak{a} \otimes M \cong \mathfrak{a}M, A \otimes M \cong M$. So we can have the exact sequence

$$\mathfrak{a}M \xrightarrow{\varphi} M \xrightarrow{\psi} (A/\mathfrak{a}) \otimes M \to 0$$

With $\operatorname{Im} \varphi = \operatorname{Ker} \psi, \operatorname{Im} \psi = (A/\mathfrak{a}) \otimes M$. Which is to say

$$\operatorname{Im} \psi = (A/\mathfrak{a}) \otimes M, \operatorname{Ker} \psi = \mathfrak{a}M$$

So by *Module Isomorphism Theorem*,

$$(A/\mathfrak{a}) \otimes M \cong M/\mathfrak{a}M$$

$\square$

Exercise 3

We follow from the hint: For any finitely generated A-module M_0, let $\mathfrak{m}$ be the maximal ideal of A and $k = M_0/\mathfrak{m}$ be the field derived from quotient. Now by *Ex-2.2*, $k \otimes_A M_0 \cong M_0/\mathfrak{m}M_0$. So by *Nakayama Lemma*, $k \otimes_A M_0 = 0$ will indicate $M_0 = 0$.

Back to the problem:

$$\begin{aligned} 0 &= k \otimes_A k \otimes_A \otimes_A M \otimes_A N \\ &\cong k \otimes_A (k \otimes_A M) \otimes_A N \\ &\cong k \otimes_A M_k \otimes_A N \\ &\cong M_k \otimes_A (k \otimes_A N) \\ &\cong M_k \otimes_A N_k \end{aligned}$$

Exercise 4

For any A-module N, consider

$$f : N \longrightarrow N'$$

which is injective. Now

$$M \text{ is flat}$$

$\Leftrightarrow$

$$f \otimes 1 : N \otimes M \longrightarrow N' \otimes M \text{ is injective}$$

$\Leftrightarrow$ (Be considering their kernel)

$$\forall i \in I, f \otimes 1 : N \otimes M_i \longrightarrow N' \otimes M_i \text{ is injective}$$

$\square$

Exercise 5

We can view $A, xA, x^2A, \ldots$ as a sequence of A-module, and $A[x]$ is their direct sum where multiplication in the A-algebra structure defined in the trivial way.

Exercise 6

Define

$$\varphi : A[x] \otimes_A M \longrightarrow M[x]$$

by mapping

$$p(x) \otimes m \longmapsto p(x) \cdot m$$

Obviously $\operatorname{Ker} \varphi = 0, \operatorname{Im} \varphi = M[x]$, so φ is an isomorphism.

Exercise 7

(i) If $a(x), b(x) \in A[x], a(x)b(x) \in \mathfrak{p}[x]$. Let $a(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0, b(x) = b_n x^n + b_{n-1}x^{n-1} + \ldots + b_1 x + b_0$, then

$$(a_n x^n + \ldots + a_1 x + a_0)(b_n x^n + \ldots + b_1 x + b_0) \in \mathfrak{p}[x]$$

$$a_n b_n \in \mathfrak{p} \Rightarrow a_n \in \mathfrak{p} \text{ or } b_n \in \mathfrak{p}$$

Suppose $a_n \in \mathfrak{p}$, then let $a'(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \ldots + a_1 x + a_0$ and we have

$$a'(x)b(x) \in \mathfrak{p}[x]$$

Repeat the process above and eventually

$$a(x) \in \mathfrak{p}[x] \text{ or } b(x) \in \mathfrak{p}[x]$$

$\square$

(ii) Not necessarily.

Let $A = k$ be a field and $\mathfrak{m} = (0)$. Obviously $\mathfrak{m}[x] = (0)$ is not a maximal ideal in $k[x]$.

Exercise 8

(i) Consider the injective A-module homomorphism

$$\varphi : R \longrightarrow R'$$

Since M is flat,

$$\varphi \otimes 1 : R \otimes_A M \longrightarrow R' \otimes_A M$$

is injective, therefore

$$\varphi \otimes 1 \otimes 1 : R \otimes_A M \otimes_A N \longrightarrow R' \otimes_A M \otimes_A N$$

is injective, i.e.

$$\varphi \otimes 1 : R \otimes_A (M \otimes_A N) \longrightarrow R' \otimes_A (M \otimes_A N)$$

is injective. So $M \otimes_A N$ is flat.

(ii) Consider a injective B-module homomorphism

$$\varphi : M \longrightarrow M'$$

Since N is a flat B-module homomorphism,

$$\varphi \otimes_B 1 : M \otimes_B N \longrightarrow M' \otimes_B N$$

is injective. To prove N is a flat A-module, we need to show

$$\varphi \otimes_A 1 : M \otimes_A N \longrightarrow M' \otimes_A N$$

is injective, i.e. $\operatorname{Ker} \varphi \otimes_A 1 = \{0\}$.

Notice $\operatorname{Ker} \varphi \otimes_A 1 \subseteq \operatorname{Ker} \varphi \otimes_B 1 = \{0\}$, we're done. $\square$

Exercise 9

Denote

$$0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

We know

$$M \cong \operatorname{Coim} g \oplus \operatorname{Ker} g \cong \operatorname{Im} g \oplus \operatorname{Ker} g$$

Be the exactness, $\operatorname{Ker} g = \operatorname{Im} f, \operatorname{Im} g = M''$, which are both finitely generated. So M is finitely generated. $\square$

Exercise 10

Since $M/\mathfrak{a}M \longrightarrow N/\mathfrak{a}N$ is surjective,

$$M \longrightarrow M/\mathfrak{a}M \longrightarrow N/\mathfrak{a}N$$

is surjective. Therefore $N = \mathfrak{a}N + u(M)$, and by *Corollary 2.7*, $u(M) = N$. $\square$

Exercise 11

(The ring multiplication here is Cartesian product!!!)

(i) When $A^m \cong A^n$: We use the method in the textbook.

Let $\mathfrak{m}$ be a maximal ideal of A, and denote the isomorphism by

$$\phi : A^m \longrightarrow A^n$$

By checking the kernel and image, we can easily see that

$$1 \otimes_A \phi : (A/\mathfrak{m}) \otimes_A A^m \longrightarrow (A/\mathfrak{m}) \otimes_A A^n$$

is an isomorphism by *Proposition 2.18.* By *Exercise 2.2* we know $(A/\mathfrak{m})^m \cong A^m/\mathfrak{m} \cong (A/\mathfrak{m}) \otimes A^m, (A/\mathfrak{m})^n \cong A^n/\mathfrak{m} \cong (A/\mathfrak{m}) \otimes A^n$ are two isomorphic linear space over the field $k = A/\mathfrak{m}$. Hence

$$\operatorname{Dim} k^m = \operatorname{Dim} k^n, m = n$$

(ii) When $\phi : A^m \longrightarrow A^n$ is surjective: Similar to *(i)*, we obtain a surjective k-linear space homomorphism by using *Proposition 2.18*

$$k^m \longrightarrow k^n$$

So $m > n$ is obtained by the property of linear space homomorphisms.

(iii) When $\phi : A^m \longrightarrow A^n$ is injective: Yes.

Note that we're no longer able to use *Proposition 2.18* to convert ring homomorphisms into linear space homomorphisms.

Suppose $m > n$, we can define A^n to be the submodule of A^m by taking the first n terms and extend the range of ϕ to A^m. Now

$$\phi(A^m) \subseteq A^n \subseteq A^m = \mathfrak{a}A^m$$

where $\mathfrak{a} = A$ is viewed as an ideal.

Since A^m is a finitely generated A-module, by *Proposition 2.4*, $\exists a_1, a_2, \ldots, a_s \in A,$ s.t.

$$\phi^s + a_1\phi^{s-1} + \ldots + a_s = 0, a_s \neq 0$$

Specifically,

$$(\phi^s + a_1\phi^{s-1} + \ldots + a_{s-1}\phi + a_s\mathrm{Id})(0, \ldots, 0, 1) = 0$$

Notice $\forall a \in A^m$, because $m > n$, the last term of $\phi(a)$ must be 0. So

$$(\phi^s + \ldots + a_{s-1}\phi)(0, \ldots, 0, 1) = (\ldots, 0)$$

But

$$a_s\mathrm{Id}(0, \ldots, 0, 1) = (\ldots, a_s)$$

Hence, it's impossible to have $(\phi^s + a_1\phi^{s-1} + \ldots + a_{s-1}\phi + a_s\mathrm{Id})(0, \ldots, 0, 1) = 0$ So $m \leq n$.

$\square$